Chú thích
PK: khóa chính | FK: khóa ngoại | UQ: giá trị duy nhất
Bảng trung gian (quan hệ n-n): room_service, booking_voucher, booking_service
Ký hiệu đường nối (từ bảng chứa FK đến bảng chứa PK):
|| đúng một | o| không hoặc một
|< một hoặc nhiều | o< không hoặc nhiều

THIẾT KẾ CƠ SỞ DỮ LIỆU (mức Logic)

Hotel Room Booking System

voucher	
PK	**voucher_code: varchar(20)**
	discount_percent: decimal(5,2)
	expiry_date: date
	is_active: boolean
FK	admin_id: varchar(10)

booking_voucher	
PK, FK	**booking_id: varchar(12)**
PK, FK	**voucher_code: varchar(20)**

payment	
PK	**payment_id: varchar(10)**
	amount: decimal(12,2)
	payment_method: varchar(20)
	payment_date: datetime
	status: varchar(20)
	refund_amount: decimal(12,2)
FK, UQ	booking_id: varchar(12)

customer	
PK	**customer_id: varchar(10)**
	sex: varchar(10)
	identity_card: varchar(12)
	address: varchar(255)
	date_of_birth: date
FK, UQ	account_id: varchar(10)

booking	
PK	**booking_id: varchar(12)**
	status: varchar(20)
	check_in_date: date
	check_out_date: date
	number_of_adult: int
	number_of_children: int
	deposit: decimal(12,2)
	VAT: decimal(12,2)
	discount: decimal(12,2)
	total_price: decimal(12,2)
FK	customer_id: varchar(10)
FK	admin_id: varchar(10)

invoice	
PK	**invoice_id: varchar(10)**
	issue_date: date
	total_amount: decimal(12,2)
FK, UQ	booking_id: varchar(12)

account	
PK	**account_id: varchar(10)**
	full_name: varchar(100)
	email: varchar(100)
	password: varchar(255)
	phone_number: varchar(15)
	failed_login_count: int
	role: varchar(20)
FK	admin_id: varchar(10)

review	
PK	**review_id: varchar(10)**
	rating: int
	content: text
	review_date: date
FK	customer_id: varchar(10)
FK	room_id: varchar(10)

booking_detail	
PK	**detail_id: varchar(10)**
	number_of_night: int
	room_price: decimal(12,2)
	sub_total: decimal(12,2)
FK	booking_id: varchar(12)
FK	room_id: varchar(10)

admin	
PK	**admin_id: varchar(10)**
FK, UQ	account_id: varchar(10)

booking_service	
PK, FK	**booking_id: varchar(12)**
PK, FK	**service_id: varchar(10)**

room	
PK	**room_id: varchar(10)**
	room_number: varchar(10)
	price_per_night: decimal(12,2)
	capacity: int
	area: decimal(8,2)
	availability: boolean
	description: text
	image: varchar(255)
FK	type_id: varchar(10)
FK	admin_id: varchar(10)

room_type	
PK	**type_id: varchar(10)**
	type_name: varchar(50)
	description: text

service	
PK	**service_id: varchar(10)**
	service_name: varchar(100)
	service_price: decimal(12,2)
FK	admin_id: varchar(10)

room_service	
PK, FK	**room_id: varchar(10)**
PK, FK	**service_id: varchar(10)**